The Bhati–Singh weak curves are subfield curves: the factor base never leaves `J_C0(F_q)`

2026-10-12. Conductor T-2193. **Structural stage diagnostic on toy fields.** No speedup is claimed, no sub-rho claim, nothing about deployed curves. Protocol frozen before the first run: PROTOCOL.md. Run `20261013T0055Z`: records, environment, run report. This report as PDF: REPORT.pdf.

Question

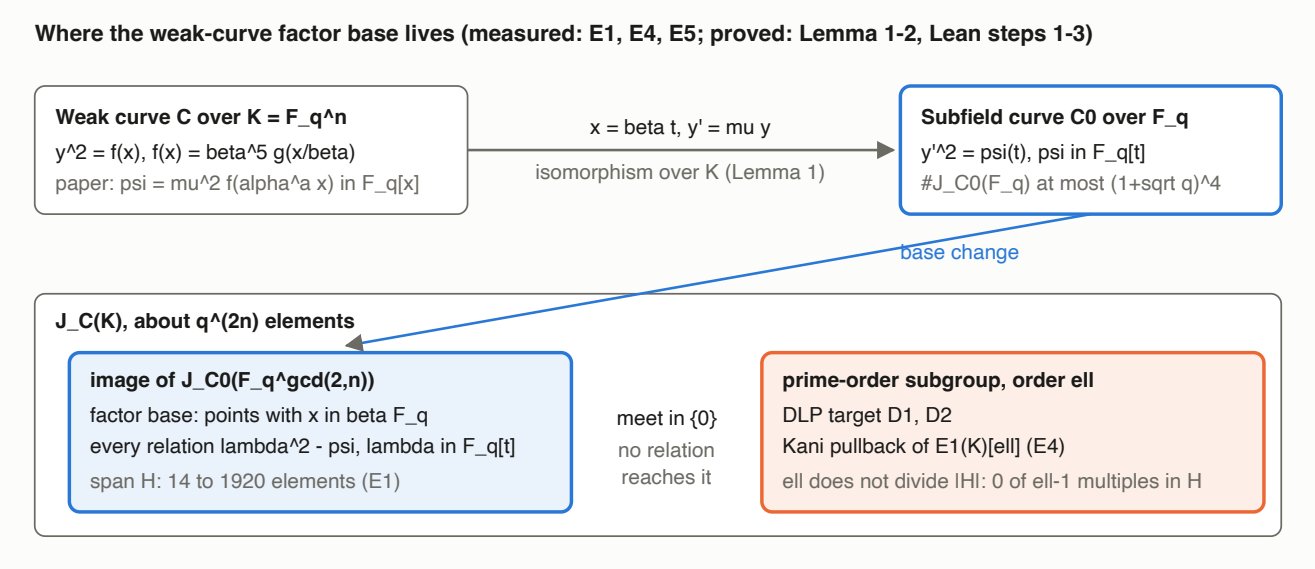
Bhati and Singh (*A new perspective on the decomposition in the Jacobian of small genus hyperelliptic curve*, SPACE) describe a class of genus-2 curves over `K = F_{q^n}` on which index-calculus relations cost no polynomial system solve. Two questions were put to it:

1. Does the relation step reach the large prime-order subgroup of `J_C(K)`?
2. If such a curve is built with a (2,2)-split Jacobian and the DLP is pulled back from an elliptic factor ("Kani descent"), does anything change? What about other factor-base lines?

Verdict

1. **No.** On a weak curve the paper's factor base generates a subgroup `H` inside the image of `J_C0(F_q)` (n odd) or `J_C0(F_{q²})` (n even), where `C0` is a curve over `F_q`. Proved below (Lemmas 1–2, Proposition 3), confirmed on the paper's own printed examples (E5) and on 15 toy weak curves (E1). Every prime factor of `|H|` is at most `(1+√q)^4`.
2. **No.** The Kani pullback `E1(K) → J_C(K)` is executable and is a homomorphism on every tested pair, but its image of prime order ℓ meets `H` only in 0 (E4). Lines other than the weak one span the group and give no free relations (E2); curves with several weak lines only reach a `(1+√q)^4`-smooth part of the group (E3b).

The trade is exact on these inputs: a line gives free relations if and only if its span collapses.



Derivation

Notation as in the paper: `C : y² = f(x)`, `f` monic quintic over `K`, `K* = <α>`, `d = (q^n − 1)/(q − 1)`, twist `f~(x) = f(α^a x)`, `μ~ = α^l`, `ψ = μ~² f~`, factor base `F~ = {(P) − (∞) : P ∈ C~(K), x(P) ∈ F_q}`.

Lemma 1 (normal form; derived, Lean step 1, certificate `IDC1h81f0f7e9f3a5f2da`). The paper's condition (16), `t_0 ≡ … ≡ t_5 (mod d)`, holds for some `a` exactly when `f(x) = β⁵ g(x/β)` with `g ∈ F_q[x]` monic and `β = α^a`. In that case `ψ ∈ F_q[x]` and `(x, y) ↦ (x, μ~ y)` is an isomorphism over `K` from `C~ : y² = f~(x)` to `C0 : y'² = ψ(x)`.

Proof. The coefficient of `x^i` in `μ~² f(α^a x)` is `α^{2l + j_i + ia}`; it lies in `F_q` for all `i ≤ 5` (with `j_5 = 0`) iff `f_i β^{i−5} ∈ F_q`, i.e. `f(βt) = β⁵ g(t)`. Then `(μ~ y)² = μ~² f~(x) = ψ(x)`. ∎

Lemma 2 (rational points; derived, Lean step 2). Every point of the factor base has, in the `C0` model, `x ∈ F_q` and `y'² = ψ(x) ∈ F_q`, so `y' ∈ F_{q²} ∩ K = F_{q^e}` with `e = gcd(2, n)`. Every relation point of `H~ = λ~² − ψ`, `λ~ ∈ F_q[x]`, has `y' = λ~(x) ∈ F_q`.

Proposition 3 (collapse; derived, step 3 in Lean, base change of Jacobians not formalized). The subgroup `H ⊂ J_C(K)` generated by the factor base lies in the image of `J_C0(F_{q^e})`. Hence `|H|` divides `#J_C0(F_{q^e})`, and since `#J_C0(F_{q²}) = #J_C0(F_q) · #J_C0^tw(F_q)`, every prime factor of `|H|` is at most `(1 + √q)^4`.

Corollary. If D1 has prime order ℓ > (1+√q)^4, then <D1> ∩ H = 0: no relation among factor-base elements carries information modulo ℓ, and no element a·D1 + b·D2 ≠ 0 decomposes over the factor base, by Nagao's method or any other. If instead ℓ ≤ (1+√q)^4, Pollard rho in <D1> already costs at most about (1 + √q)² ≈ q group operations, which is not more than the paper's relation collection over a factor base of size O(q).

The paper states that the individual-logarithm step "still needs" Nagao's decomposition. On its weak curves that step has no solutions for a target of large prime order. The paper's experiments exhibit single relations only; they do not compute a discrete logarithm.

The same argument covers any line L = βF_q + b on which c · f(βt + b) has coefficients in F_q for some c ∈ K* (a "weak line"), and any union of such lines: each contributes a subgroup of some F_q-form of the curve.

Lean: lean/SubfieldCollapse.lean, toolchain `leanprover/lean4:v4.34.1`, no Mathlib, `lake build` log in lean/build.log. `#print axioms` reports no axioms for `point_map`, `relation_point_rational`, `span_le_image`, `not_in_span_of_not_in_image`.

Report statement	Lean declaration	Status
Lemma 1, map to C0	`point_map`	proved for any commutative multiplication
Lemma 2, relation points rational	`relation_point_rational`	proved
Prop. 3, span inside the image	`span_le_image`	proved for any map preserving 0, +, −
Corollary, target outside the span	`not_in_span_of_not_in_image`	proved
φ : J_C0(F_{q^e}) → J_C(K) exists and sends a rational point class to (P) − (∞)	none	**not formalized** (Jacobians of curves are not available without Mathlib; standard base change)
#J_C0(F_q) ≤ (1+√q)^4 (Weil)	none	**not formalized**

Measurements (run 20261013T0055Z, all exact counts)

Tool: weak_curve_lab.rs, dependency-free Rust, Cantor arithmetic on Mumford divisors over F_{q^n} by log tables; subgroups by exact coset extension. #J(K) from point counts over K and K², and for weak curves also from det(1 − πⁿ) of the F_q L-polynomial: the two agreed on all 15 weak curves and all 6 Kani curves. Group law self-test (#J · D = 0, associativity, 20 random triples per curve) passed on all 30 E1 curves.

E5: the paper's printed examples

Example	q	n	μ~² f~ = ψ with ψ over F_q	points on C~	μ~·y ∈ F_q	μ~·y = ±λ~(x)
1	1354523531	7	yes (6 of 6 coefficients)	5 of 5	5 of 5	5 of 5
2	13545235267	4	yes	5 of 5	5 of 5	5 of 5

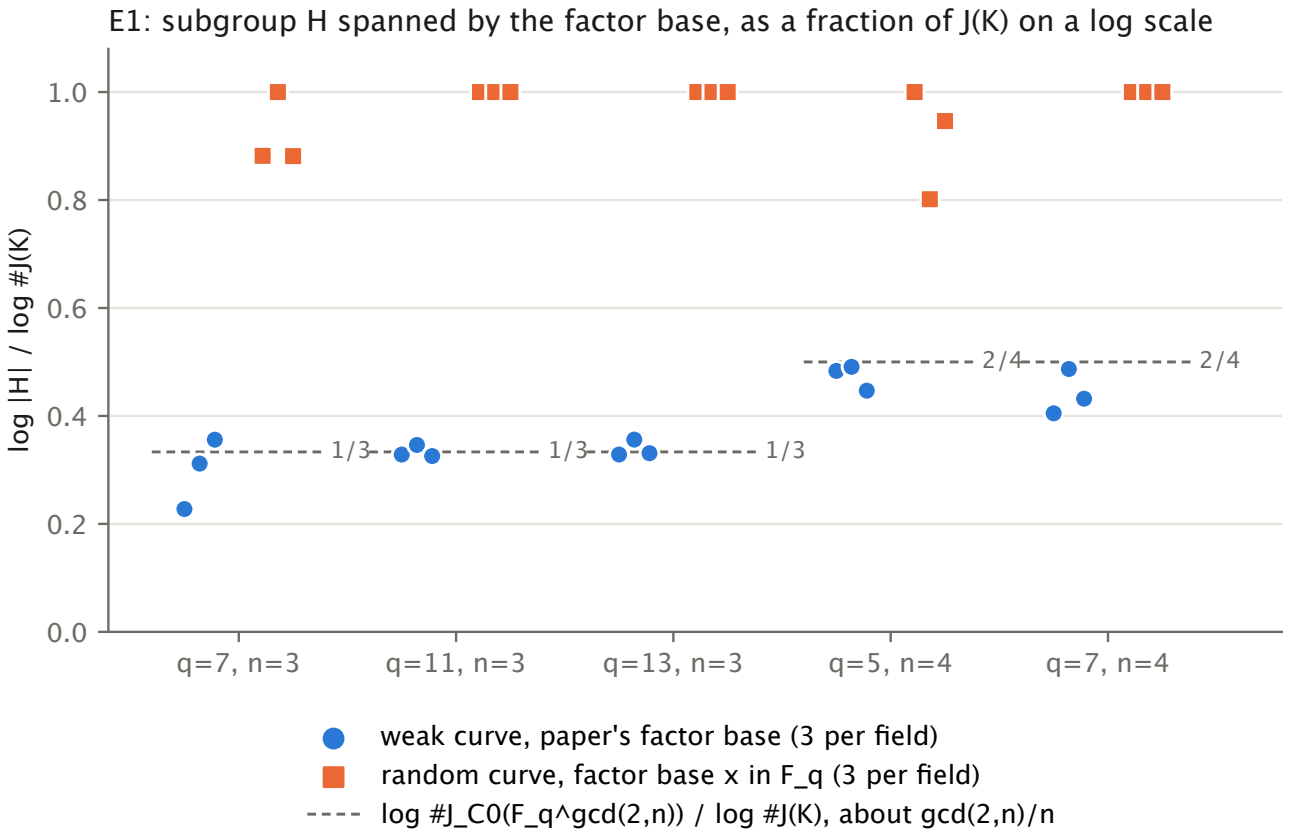
All ten published relation points are F_q-rational points of C0.

E1: span of the factor base, weak against random

Three weak curves and three random curves per field. sub is #J_C0(F_q) for n = 3 and #J_C0(F_{q²}) for n = 4.

(q, n)	#J(K) (weak)	size of H (weak)	sub	largest prime of #J(K)	largest prime of size of H	(1+√q)^4	index of H on the 3 random curves
(7, 3)	108892, 127413, 118528	14, 39, 64	28, 39, 64	3889, 13, 463	7, 13, 2	176.7	4, 1, 4
(11, 3)	1734208, 1856068, 1751868	112, 148, 108	112, 148, 108	79, 12541, 5407	7, 37, 3	347.2	1, 1, 1
(13, 3)	4739904, 4652073, 4928036	156, 237, 164	156, 237, 164	211, 727, 199	13, 79, 41	449.9	1, 1, 1
(5, 4)	392656, 438805, 456976	506, 589, 338	1012, 589, 676	97, 149, 13	23, 31, 13	109.7	1, 13, 2
(7, 4)	6137600, 5529600, 5831936	560, 1920, 836	2240, 3840, 3344	137, 5, 109	7, 5, 19	176.7	1, 1, 1

On all 15 weak curves |H| divides sub, every element of H has all its C0-model Mumford coordinates in F_{q^e}, and the paper's condition (16) holds (an a was found). On all 15 random curves condition (16) fails and H has index 1 to 13 in J(K); the three cases above index 2 have a factor base of one point. P1 and P2 hold; the falsifier did not fire.

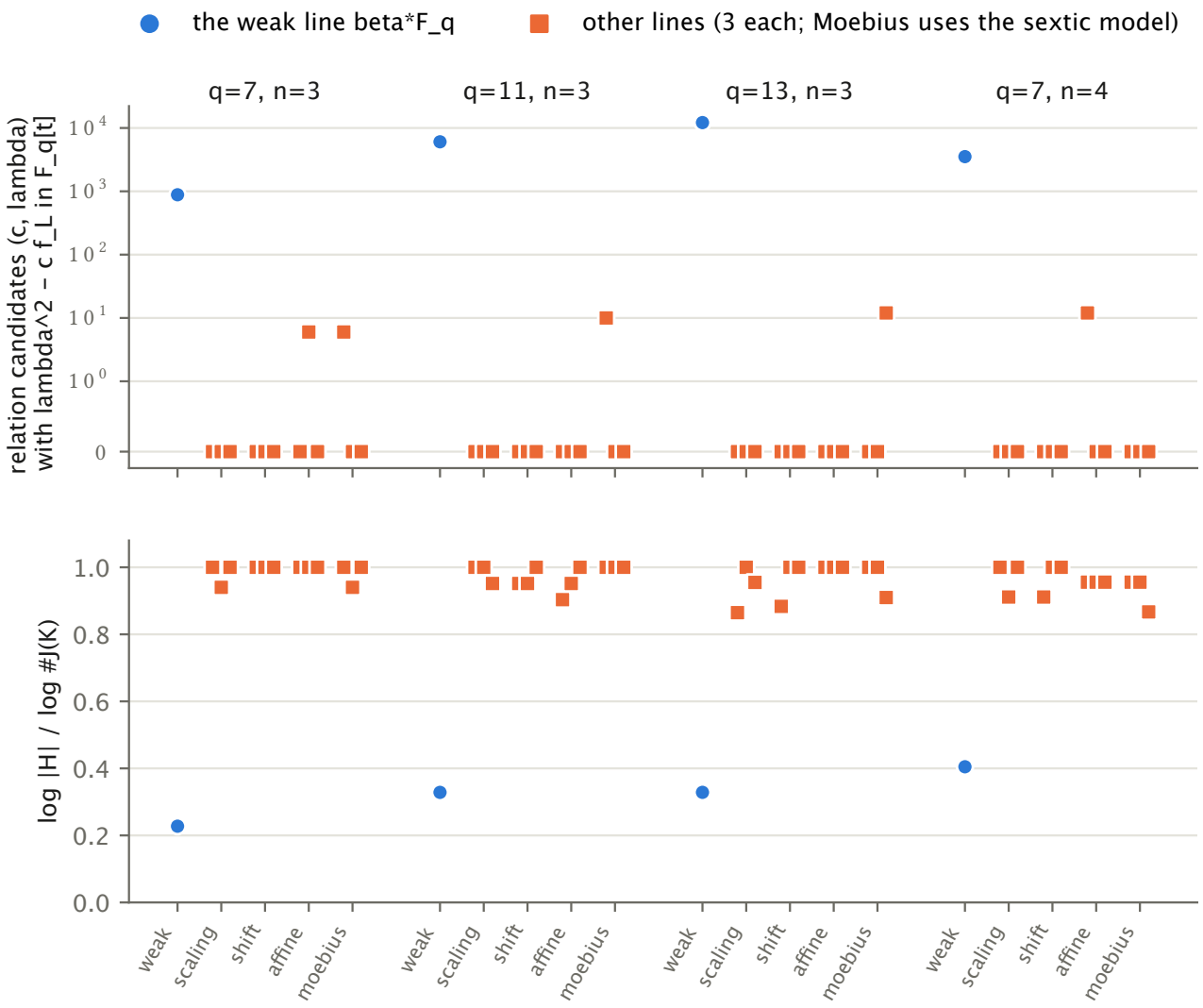


E2: other factor-base lines on a weak curve

First weak curve of E1 in each field. "Candidates" counts pairs `(c, λ)`, `c` a square, `deg λ = 2` (quintic model, affine lines) or `3` (sextic model, Möbius lines), with `λ² − c·f_L ∈ F_q[t]`. Constant μ only; the two models are not like-for-like. Each cell lists the three lines as candidates / index of `H`.

`(q, n)`	weak line `βF_q`	3 scalings `β'F_q`	3 shifts `βF_q + b`	3 generic affine	3 Möbius
(7, 3)	882 / 7778	0/1, 0/2, 0/1	0/1, 0/1, 0/1	0/1, 6/1, 0/1	6/1, 0/2, 0/1
(11, 3)	6050 / 15484	0/1, 0/1, 0/2	0/2, 0/2, 0/1	0/4, 0/2, 0/1	10/1, 0/1, 0/1
(13, 3)	12168 / 30384	0/8, 0/1, 0/2	0/6, 0/1, 0/1	0/1, 0/1, 0/1	0/1, 0/1, 12/4
(7, 4)	3528 / 10960	0/1, 0/4, 0/1	0/4, 0/1, 0/1	12/2, 0/2, 0/2	0/2, 0/2, 0/8

The weak line's count equals `(q−1)² q² / 2` for `n = 3` and four times that for `n = 4` (derived: `λ ∈ F_q[t]`, and for even `n` also `√ε · F_q[t]` and both square classes of `c`). On the 48 other lines the count is 0 on 43 and 6 to 12 on 5, which is one orbit under `(c, λ) → (e²c, ±eλ)`, `e ∈ F_q*`; all 48 span `J(K)` up to index 8.



E3: exhaustive census of weak affine lines

All `d · q^{n−1}` affine lines of each curve.

`(q, n)`	lines	weak curve	random curve	$x^5 + c$	$x^5 + cx$
(7, 3)	2793	1	0	1	1
(11, 3)	16093	1	0	1	1
(13, 3)	30927	1	0	1	1
(5, 4)	19500	1	0	not run (char 5)	0
(7, 4)	137200	2	0	0	0
(3, 4)	1080	1	0	0	0

A random curve has no weak line in any field. For `n = 3` every sampled $x^5 + c$ and $x^5 + cx$ is weak (derived: `gcd(5, d) = gcd(4, d) = 1`, so $\beta^5 \in c \cdot F_q^*$ is solvable). **Unexplained:** the (7, 4) weak curve has a second weak line that is not a scaling line; its span was not computed. A candidate explanation is an extra affine symmetry of that `g`; not checked.

E3b: all weak lines of automorphism-rich curves

First random `c` for which at least one weak line exists (`tries` in the records).

curve, `(q, n)`	weak lines	Frobenius forms	per-line size of `H`	union size of `H`	`#J(K)`	largest prime, union or per line	`(1+√q)^4`
x^5+c, (3, 4)	5	1	50 ×5	2000	10000 = $2^4 \cdot 5^4$	5	55.7
x^5+c, (7, 4)	5	1	1250 ×5	1250000	6250000 = $2^4 \cdot 5^8$	5	176.7
x^5+c, (11, 5)	5	5	101, 176, 1, 131, 271	631065776 (derived: pairwise coprime, product)	25873696816 = $2^4 \cdot 11 \cdot 41 \cdot 101 \cdot 131 \cdot 271$	271	347.2

curve, (q, n)	weak lines	Frobenius forms	per-line size of H	union size of H	#J(K)	largest prime, union or per line	(1+√q)^4
x⁵+cx, (5, 4)	6	2	16, 324 ×5	1296	331776 = $2^{12}\cdot3^4$	3	109.7
x⁵+cx, (7, 4)	4	1	1024, 512, 1024, 512	2048	5308416 = $2^{16}\cdot3^4$	2	176.7
x⁵+cx, (13, 4)	2	1	14450 ×2	≥ 104401250 (capped)	835210000 = $2^4\cdot5^4\cdot17^4$	17	449.9

Here the union of weak lines can be a large part of J(K): index 5 at (3, 4) and (7, 4), index 41 at (11, 5). In every row the largest prime factor of every per-line span, and of every union that was enumerated, is below (1+√q)^4, and so is the largest prime factor of #J(K) itself: these Jacobians split into F_q-sized pieces, which is what makes several lines weak. P3 holds. For the two capped unions the bound is established for the per-line spans and for #J(K), hence for the union.

E4: (2,2)-split weak curve and the pullback from E1

C0 : y² = S(x) = x⁶ + ax⁴ + bx² + c with a root r ∈ F_q moved to infinity; E1 : v² = u³ + au² + bu + c, E2 : v² = cu³ + bu² + au + 1; C = the twist f(x) = β⁵ ψ0(x/β), β = γ². The map φ : E1(K) → J_C(K) is the pullback along (x, y) ↦ (x², y). E4 is the frozen selection (first (a, b, r), which is S = x⁶ − 1); E4x requires a, b ≥ 1 and was added after E4 was seen (supplementary).

run	(q, n)	S	#J(K) = #E1(K)·#E2(K)	φ(P+Q) = φ(P)+φ(Q)	ℓ	order of φ(P)	size of H	multiples of φ(P) in H
E4	(7, 3)	x⁶+6	117936 = 364·324	200 of 200	13	13	48 = $2^4\cdot3$	0 of 12
E4	(11, 3)	x⁶+10	1774224 = 1332·1332	199 of 199	37	37	144 = $2^4\cdot3^2$	0 of 36
E4	(13, 3)	x⁶+12	5143824 = 2268·2268	200 of 200	7	7	144 = $2^4\cdot3^2$	0 of 6
E4x	(7, 3)	x⁶+x⁴+x²+4	100440 = 310·324	200 of 200	31	31	120 = $2^3\cdot3\cdot5$	0 of 30
E4x	(11, 3)	x⁶+x⁴+x²+8	1683648 = 1332·1264	198 of 198	37	37	192 = $2^6\cdot3$	0 of 36
E4x	(13, 3)	x⁶+x⁴+x²+10	4435236 = 2106·2106	200 of 200	13	13	324 = $2^2\cdot3^4$	0 of 12

ℓ is the largest prime factor of #E1(K); pairs where a point maps to the excluded fibre u = r² were skipped, which is why two rows test 198 or 199 pairs. The transfer is executable and injective on the ℓ-part, and the transferred instance lies outside the factor-base span in all six cases. P4 holds.

Traces of π^n reachable from an F_q-curve, against all integer traces in the Hasse interval over K (measured counts): 7 of 75 at (7, 3), 13 of 145 at (11, 3), 11 of 187 at (13, 3).

What the Kani route cannot do (derived)

- **Compatibility.** A weak curve is a base change, so its elliptic factors are defined over F_q (or F_{q²}). Only elliptic curves over K isogenous to a subfield curve are reachable: at most 4√q + 1 of about 4√Q Frobenius traces, Q = q^n, a fraction about q^{−(n−1)/2}.
- **Factor base.** Proposition 3 applies unchanged to the transferred instance (E4).
- **Direction.** The transfer goes from a group of size about Q to one of size about Q². Rho on E(K) costs about Q^{1/2}, on J_C(K) about Q; genus-2 index calculus over K has the same exponent as rho (2 − 2/g = 1 at g = 2).
- This is the case "Jacobian embedding plus base change" of the transfer conservation note docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md.

Scope and limits

- Toy fields, q ≤ 13, n ∈ {3, 4, 5}, genus 2, odd characteristic. The derivation is uniform in q, n; E5 checks it at the paper's 31- and 34-bit q.
- Non-weak lines were probed with constant μ and bounded deg λ only. The Joux–Vitse system with deg μ ≥ 1 was not run; E2 says nothing about how many relations it finds there.
- The characteristic-2 variant (Section 6 of the paper) was not tested. The same substitution y' = μ~ y applies to its μ~² f~ term; the paper already reports too few relations there.
- Kani splittings of degree above 2 were not built. The compatibility and factor-base arguments do not depend on the degree.
- Nothing here bounds the cost of the HCDLP on these curves by other methods. A base-changed curve over F_{q^n} is the classical setting for Weil-descent attacks; that is outside this study.
- Unresolved: the second weak line at (7, 4) in E3.

Graphs checked and left unchanged

No canonical graph changes: docs/index-calculus-scoreboard.html, docs/ic/progress-timeline.json and docs/ic/boundary_targets.json carry measured ECDLP rungs, and this study adds none. No timing was taken, so benchmarks/ has no entry.

Reproduce

```
rustc -O weak_curve_lab.rs && ./weak_curve_lab all && ./weak_curve_lab e4x
python3 reports/plot.py results/run-20261013T0055Z.jsonl reports/run-20261013T0055Z   # layout only
cd lean && lake build
cargo run --release --example identity_certificate_weak_twist
```